

Speaker

DO

OUTPUT

SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses



Door chime

DETECT
Button and
chime setting



DECIDE
Which chime
should play?



DO
Play the notes



Tiny fanfare machine

DETECT
Count
achievement-
button presses



DECIDE
Has the
target been
reached?



DO
Play
triumphant
notes



Cone
Pushes the air

Coil and magnet
Create
movement

Terminals
Connect
through 220
ohms

Code

● Raspberry Pi

```
from gpiozero import TonalBuzzer  
speaker = TonalBuzzer(17)  
speaker.play("E4")
```

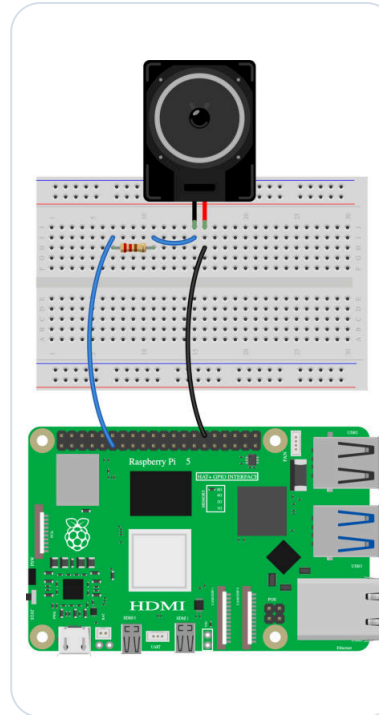
● Raspberry Pi Pico

```
from picozero import Speaker  
speaker = Speaker(16)  
speaker.play("E4", duration=1)
```

For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

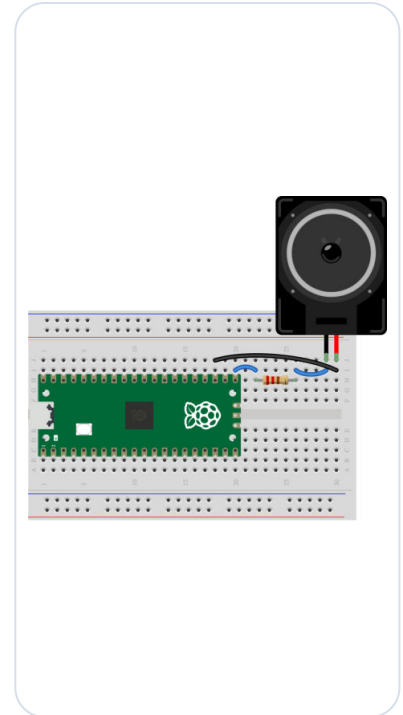
Wiring

RASPBERRY PI 5



GPIO17 → 220 ohms → speaker → **GND**

PICO



GP16 → 220 ohms → speaker → **GND**